A Theoretical and Empirical Comparison of Rule-Based, Statistical, and Neural POS Tagging

Mini Project / Term Project

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- **Repository:** https://github.com/vvkmlt/POS-Tagging-Project

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This notebook implements and benchmarks three families of Part-of-Speech (POS) taggers on the Penn Treebank sample distributed with NLTK:

Family	Implementation	Learning signal
Rule-based	Hand-written regex/lexicon tagger	Human linguistic knowledge
Rule-based (learned rules)	Brill transformation-based learning	Error-driven rule induction
Statistical	Bigram HMM with Viterbi decoding	Transition and emission counts
Statistical	Trigram tagger with back-off	N-gram counts
Neural	Character-aware BiLSTM	Learned distributed representations

All models are trained on the same split and evaluated on the same held-out test sentences.

1. Introduction and Objectives

POS tagging assigns a grammatical category (noun, verb, determiner, ...) to every token in a sentence. It is a foundational sequence-labelling task that feeds parsing, named-entity recognition, machine translation, and information extraction.

The project answers three questions:

1. **How large is the accuracy gap** between hand-crafted rules, probabilistic sequence models, and neural sequence models on the same corpus?

2. **How do the approaches behave on out-of-vocabulary (OOV) words**, where memorisation and hand-written rules are weakest?
3. **What are the engineering trade-offs** in training cost, inference speed, and interpretability?

Deliverables covered by this notebook

- Complete, reproducible Jupyter notebook with all experiments.
- Tables, figures, error analyses, and textual discussion for the PDF report.
- Saved artifacts (`reports/results.json` , `reports/model_comparison.csv` , `assets/figures/*.png`) used to build the report and the screenshot PDF.

2. Experimental Setup

The following choices keep the comparison fair:

- **Corpus:** `nltk.corpus.treebank` (Penn Treebank sample, 3,914 sentences, PTB tagset).
- **Split:** sentence-level 80/10/10 train/validation/test with a fixed seed (42). Validation data is used only for neural early stopping; it is never used to fit the rule-based or statistical taggers.
- **Evaluation:** every tagger receives raw token sequences and must return one tag per token.
- **Metrics:** token accuracy, macro F1, weighted F1, OOV accuracy, inference throughput, and training time.
- **Hardware:** timings are measured on the CPU of the machine running this notebook.

In [1]:
```
import json
import math
import platform
import sys
import time
from pathlib import Path

import matplotlib.pyplot as plt
import numpy as np
import pandas as pd
import torch
from sklearn.metrics import classification_report


def find_project_root() -> Path:
    for candidate in [Path.cwd(), *Path.cwd().parents]:
        if (candidate / "src" / "data.py").exists():
            return candidate
    raise FileNotFoundError("Run this notebook from inside the POS tagging project
```

```
ROOT = find_project_root()
if str(ROOT) not in sys.path:
    sys.path.insert(0, str(ROOT))

FIG_DIR = ROOT / "assets" / "figures"
REPORT_DIR = ROOT / "reports"
for directory in (FIG_DIR, REPORT_DIR):
    directory.mkdir(parents=True, exist_ok=True)

SEED = 42
np.random.seed(SEED)
torch.manual_seed(SEED)

plt.rcParams.update(
    {
        "figure.dpi": 120,
        "savefig.dpi": 150,
        "savefig.bbox": "tight",
        "axes.spines.top": False,
        "axes.spines.right": False,
        "axes.grid": True,
        "grid.alpha": 0.25,
        "font.size": 10,
    }
)

COLORS = ["#4C72B0", "#DD8452", "#55A868", "#C44E52", "#8172B3", "#937860"]

from src.data import (
    count_tokens,
    load_treebank,
    oov_words,
    split_sentences,
    tagset,
    tags_of,
    tokens_of,
)
from src.evaluate import (
    error_examples,
    evaluate_tagger,
    per_tag_frame,
    results_frame,
)
from src.neural import NeuralTagger, build_vocabs, train_bilstm
from src.rule_based import REGEX_RULES, BrillRuleTagger, RegexRuleTagger
from src.statistical import HMMTagger, NgramBackoffTagger

print("Project root:", ROOT)
print("Python:", platform.python_version())
print("PyTorch:", torch.__version__)
print("Device: CPU")
print("Processor:", platform.processor() or platform.machine())
```

```
Project root: E:\TA
Python: 3.14.3
PyTorch: 2.9.0+cpu
Device: CPU
Processor: Intel64 Family 6 Model 186 Stepping 2, GenuineIntel
```

3. Dataset Preparation and Exploratory Data Analysis

The Penn Treebank sample is already tokenised and tagged with the 45-tag PTB tagset. We keep the original tagset so that the results are directly comparable across all five models.

The split is performed at the **sentence** level to avoid leaking fragments of the same sentence into training and testing. The test partition is used once, after all models are fitted.

In [2]:
```
sentences = load_treebank()
split = split_sentences(sentences, train_frac=0.8, val_frac=0.1, seed=SEED)

summary = pd.DataFrame(split.summary().items(), columns=["Property", "Value"]).set_
display(summary)

train_tags = tagset(split.train)
test_tags = tagset(split.test)
print(f"Tags observed in training: {len(train_tags)}")
print(f"Tags observed in test:     {len(test_tags)}")
print("Training tagset:", ", ".join(train_tags))
```

Property	Value
seed	42
train_sentences	3131
val_sentences	391
test_sentences	392
train_tokens	81021
val_tokens	9660
test_tokens	9995

```
Tags observed in training: 46
Tags observed in test:     41
Training tagset: #, $, '', ,, -LRB-, -NONE-, -RRB-, ., :, CC, CD, DT, EX, FW, IN, J
J, JJR, JJS, LS, MD, NN, NNP, NNPS, NNS, PDT, POS, PRP, PRP$, RB, RBR, RBS, RP, SYM,
TO, UH, VB, VBD, VBG, VBN, VBP, VBZ, WDT, WP, WP$, WRB, ``
```

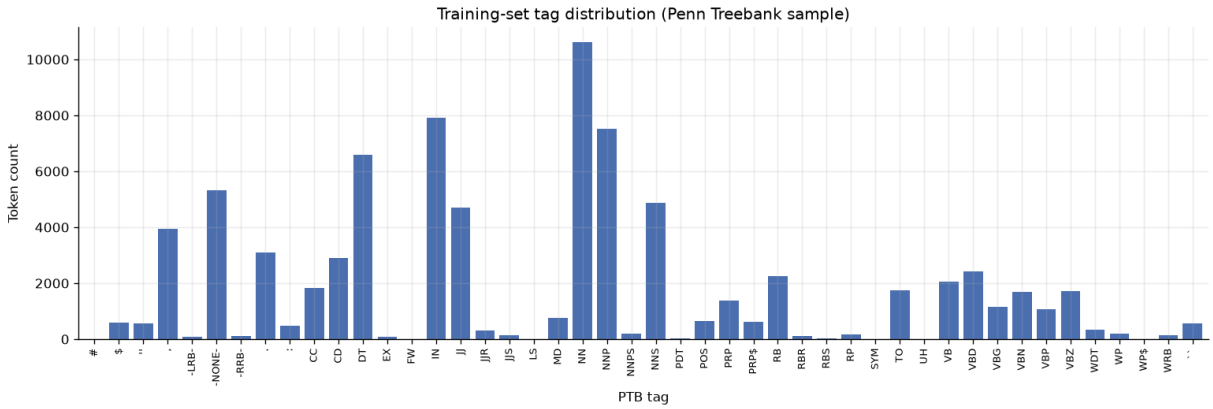
In [3]:
```
all_tagged = split.train + split.val + split.test
tag_counts = pd.Series(tags_of(all_tagged)).value_counts()
tag_table = pd.DataFrame(
    {
```

```
        "Train": pd.Series(tags_of(split.train)).value_counts(),
        "Validation": pd.Series(tags_of(split.val)).value_counts(),
        "Test": pd.Series(tags_of(split.test)).value_counts(),
    }
).fillna(0).astype(int)
tag_table["Share of Test Tokens"] = (tag_table["Test"] / tag_table["Test"].sum()).r
tag_table.index.name = "Tag"
display(tag_table.head(20))

fig, ax = plt.subplots(figsize=(13, 4.5))
tag_table["Train"].plot.bar(ax=ax, color=COLORS[0], width=0.8)
ax.set_title("Training-set tag distribution (Penn Treebank sample)")
ax.set_xlabel("PTB tag")
ax.set_ylabel("Token count")
ax.tick_params(axis="x", rotation=90, labelsize=8)
fig.tight_layout()
fig.savefig(FIG_DIR / "fig_tag_distribution.png")
plt.show()

print(f"Most frequent tag: {tag_counts.index[0]} ({tag_counts.iloc[0]} tokens)")
print(f"Rarest tag:       {tag_counts.index[-1]} ({tag_counts.iloc[-1]} tokens)")
```

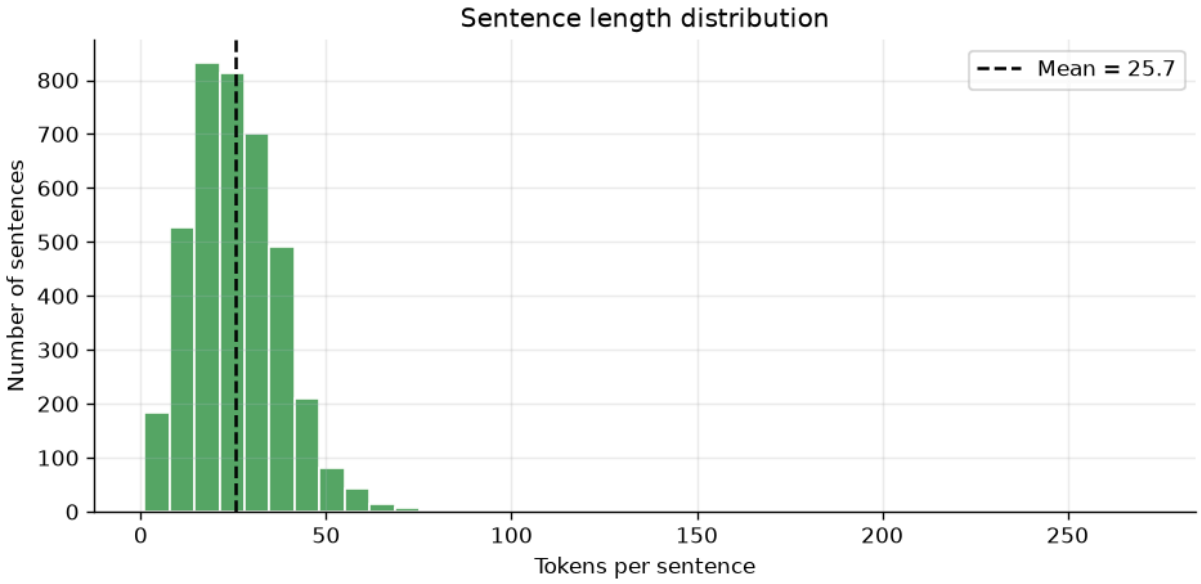
Tag	Train	Validation	Test	Share of Test Tokens
#	13	0	3	0.0003
$	586	56	82	0.0082
''	559	68	67	0.0067
,	3948	443	495	0.0495
-LRB-	96	10	14	0.0014
-NONE-	5333	614	645	0.0645
-RRB-	99	13	14	0.0014
.	3101	386	387	0.0387
:	478	41	44	0.0044
CC	1831	225	209	0.0209
CD	2899	295	352	0.0352
DT	6582	758	825	0.0825
EX	71	11	6	0.0006
FW	4	0	0	0.0000
IN	7924	947	986	0.0986
JJ	4696	555	583	0.0583
JJR	298	43	40	0.0040
JJS	147	16	19	0.0019
LS	10	3	0	0.0000
MD	763	83	81	0.0081




```
Most frequent tag: NN (13166 tokens)
Rarest tag:        SYM (1 tokens)
```

In [4]:
```
sentence_lengths = np.array([len(sentence) for sentence in all_tagged])
fig, ax = plt.subplots(figsize=(8, 4))
ax.hist(sentence_lengths, bins=40, color=COLORS[2], edgecolor="white")
ax.set_title("Sentence length distribution")
ax.set_xlabel("Tokens per sentence")
ax.set_ylabel("Number of sentences")
ax.axvline(sentence_lengths.mean(), color="black", linestyle="--", label=f"Mean = {
ax.legend()
fig.tight_layout()
fig.savefig(FIG_DIR / "fig_sentence_lengths.png")
plt.show()

print(f"Sentences: {len(all_tagged):,}")
print(f"Tokens:    {len(tokens_of(all_tagged)):,}")
print(f"Longest sentence: {sentence_lengths.max()} tokens")
```




```
Sentences: 3,914
Tokens:    100,676
Longest sentence: 271 tokens
```

In [5]:
```
oov = oov_words(split.train, split.test)
test_tokens = tokens_of(split.test)
test_oov_mask = np.array([token in oov for token in test_tokens])

oov_frame = pd.DataFrame(
    {
        "Quantity": [
            "Test tokens",
            "OOV tokens",
            "OOV rate (tokens)",
            "Distinct test word types",
            "Distinct OOV word types",
            "OOV rate (types)",
        ],
        "Value": [
            len(test_tokens),
            int(test_oov_mask.sum()),
            f"{test_oov_mask.mean():.2%}",
```

```
            len(set(test_tokens)),
            len(oov),
            f"{len(oov) / len(set(test_tokens)):.2%}",
        ],
    }
).set_index("Quantity")
display(oov_frame)

sample_oov = pd.Series(sorted(oov)).sample(15, random_state=SEED)
print("Sample OOV words:", ", ".join(sample_oov))
```

Quantity	Value
Test tokens	9995
OOV tokens	721
OOV rate (tokens)	7.21%
Distinct test word types	3118
Distinct OOV word types	704
OOV rate (types)	22.58%

```
Sample OOV words: Waste, 40,000, 374.20, Laser, ambassador, Strait, Philip, asset-va
luation, Toys, Alexander, ECONOMIC, Filling, directed, erroneous, Norwick
```

In [6]:
```
pd.set_option("display.max_colwidth", 140)
sample_row = pd.DataFrame(
    [
        {
            "Sentence (tokens)": " ".join(word for word, _ in sentence),
            "Gold tags": " ".join(tag for _, tag in sentence),
        }
        for sentence in split.test[:5]
    ]
)
display(sample_row)
pd.reset_option("display.max_colwidth")
```

	Sentence (tokens)	Gold tags
0	This species of congressional action is predicated *-50 on an interpretation of the appropriations clause that *T*-1 is erroneous and un...	DT NN IN JJ NN VBZ VBN -NONE- IN DT NN IN DT NNS NN WDT -NONE- VBZ JJ CC JJ .
1	If , when trading resumes *T*-1 , the S&P futures fall 30 points from the previous day 's close , a one-hour trading halt takes effect .	IN , WRB NN VBZ -NONE- , DT NNP NNS VBP CD NNS IN DT JJ NN POS NN , DT JJ NN NN VBZ NN .
2	Oshkosh Truck Corp. , Oshkosh , Wis. , estimated 0 earnings for its fourth quarter ended Sept. 30 fell 50 % to 75 % below the year-earli...	NNP NNP NNP , NNP , NNP , VBN -NONE- NNS IN PRP$ JJ NN VBD NNP CD VBD CD NN TO CD NN IN DT JJ $ CD CD -NONE- , CC CD NNS DT NN .
3	Stadiums tend *-1 to redistribute existing wealth within a community , not create more of it .	NNS VBP -NONE- TO VB VBG NN IN DT NN , RB VB JJR IN PRP .
4	Western Union indeed wanted *-1 to get into the telephone business .	NNP NNP RB VBD -NONE- TO VB IN DT NN NN .

Evaluation protocol

For each model we compute:

- **Accuracy** = correctly tagged tokens / all test tokens.
- **Macro F1** = unweighted mean of per-tag F1, which exposes weakness on rare tags.
- **Weighted F1** = support-weighted mean of per-tag F1 (comparable to accuracy but penalising systematic confusions).
- **OOV accuracy** = accuracy restricted to test tokens that never occurred in training.
- **Inference throughput** = test tokens per second, tagging one sentence at a time on CPU.
- **Training time** = wall-clock time to fit the model (zero for the hand-written tagger).

Training time is measured separately from inference time; neural inference includes the PyTorch forward pass and tensor construction per sentence, which is the realistic deployment API here.

4. Approach 1: Rule-Based Tagging

4.1 Hand-written regex and lexicon rules

A rule-based tagger encodes linguistic knowledge directly. We use an ordered cascade:

1. Punctuation and symbol lookup.
2. Irregular-form lexicon (e.g., *was* -> `VBD`, *children* -> `NNS`).
3. Closed-class word lists (pronouns, determiners, prepositions, modals, ...).

4. Morphological regex rules (e.g., `-ing` -> `VBG`, `-ed` -> `VBD`, `-ly` -> `RB`, capitalised token -> `NNP`, trailing `-s` -> `NNS`).
5. Default tag `NN` when no rule fires.

The cascade is ordered from most to least specific. This model has **no learned parameters**, so its training time is zero, and every decision can be inspected by a human.

Limitations: English morphology and syntax are highly ambiguous. A single suffix is not a reliable tag predictor (*running* can be `VBG` or `NN`; *fast* can be `JJ`, `RB`, or `NN`), and hand-written rules cannot resolve context-dependent ambiguity.

In [7]:
```
regex_tagger = RegexRuleTagger()

rule_table = pd.DataFrame(REGEX_RULES, columns=["Regex pattern", "Assigned tag"])
display(rule_table)

demo_tokens = "The quick brown foxes were running through the old houses .".split()
demo_frame = pd.DataFrame(
    {
        "Token": demo_tokens,
        "Predicted tag": regex_tagger.tag(demo_tokens),
    }
)
display(demo_frame)

start = time.perf_counter()
regex_result = evaluate_tagger(regex_tagger.name, regex_tagger.tag, split.test, oov
regex_train_seconds = 0.0
regex_eval_seconds = time.perf_counter() - start

display(pd.DataFrame([regex_result.metrics()]).round(4))
print(classification_report(regex_result.gold, regex_result.predictions, zero_divis
```

	Regex pattern	Assigned tag
0	^[.,;:!?]+$	.
1	^[-A-Za-z0-9/]+\.$	NNP
2	^[A-Z]{2,}$	NNP
3	^[A-Z][a-z]+$	NNP
4	^-?\$?\d[\d,.]*$	CD
5	^[A-Za-z]+ly$	RB
6	^[A-Za-z]+ing$	VBG
7	^[A-Za-z]+ed$	VBD
8	^[A-Za-z]+(tion|sion|ment|ness|ity|ance|ence|s...	NN
9	^[A-Za-z]+(ous|ful|able|ible|ive|al|ic|ary|les...	JJ
10	^[A-Za-z]+(ss|us|is)$	NN
11	^[A-Za-z]+(s|es)$	NNS

	Token	Predicted tag
0	The	DT
1	quick	NN
2	brown	NN
3	foxes	NNS
4	were	VBD
5	running	VBG
6	through	IN
7	the	DT
8	old	NN
9	houses	NNS
10	.	.

	Model	Accuracy	Macro F1	Weighted F1	OOV Accuracy	Tokens	OOV Tokens	Inference Seconds	Tokens/Second
0	Rule-Based (Regex)	0.7388	0.5322	0.6982	0.6241	9995	721	0.0103	968836.332

	precision	recall	f1-score	support
#	1.00	1.00	1.00	3
$	1.00	1.00	1.00	82
''	1.00	0.99	0.99	67
,	1.00	1.00	1.00	495
-LRB-	0.00	0.00	0.00	14
-NONE-	0.00	0.00	0.00	645
-RRB-	0.00	0.00	0.00	14
.	1.00	1.00	1.00	387
:	1.00	0.59	0.74	44
CC	0.88	1.00	0.94	209
CD	0.73	0.68	0.70	352
DT	0.89	1.00	0.94	825
EX	0.00	0.00	0.00	6
IN	0.94	0.85	0.90	986
JJ	0.80	0.29	0.42	583
JJR	0.00	0.00	0.00	40
JJS	0.00	0.00	0.00	19
MD	0.95	0.94	0.94	81
NN	0.41	0.88	0.56	1326
NNP	0.86	0.93	0.89	900
NNPS	0.00	0.00	0.00	27
NNS	0.81	0.95	0.88	586
PDT	0.00	0.00	0.00	4
POS	0.00	0.00	0.00	83
PRP	0.94	0.99	0.96	161
PRP$	0.99	1.00	0.99	77
RB	0.81	0.65	0.72	308
RBR	0.00	0.00	0.00	10
RP	0.00	0.00	0.00	26
TO	1.00	1.00	1.00	212
UH	0.00	0.00	0.00	0
VB	0.73	0.23	0.35	250
VBD	0.57	0.90	0.70	300
VBG	0.64	0.96	0.77	140
VBN	1.00	0.06	0.12	223
VBP	1.00	0.27	0.43	135
VBZ	1.00	0.62	0.76	207
WDT	0.00	0.00	0.00	49
WP	0.46	1.00	0.63	23
WP$	0.00	0.00	0.00	3
WRB	1.00	1.00	1.00	26
``	1.00	1.00	1.00	67
accuracy			0.74	9995
macro avg	0.58	0.54	0.53	9995
weighted avg	0.73	0.74	0.70	9995

4.2 Brill transformation-based learning (TBL)

Brill tagging still produces a symbolic, human-readable rule set, but the rules are **learned from data** rather than written by hand.

1. Start with a unigram tagger as the baseline (most frequent tag per word).
2. Compare the baseline output with the gold tags on the training set.
3. Enumerate candidate transformations of the form *"change tag X to tag Y when context condition C holds"*.
4. Greedily apply the transformation that removes the most errors (net error reduction), then repeat until the rule budget is exhausted.

The result is a compact cascade of contextual rules, e.g. *"change* `NN` *to* `VBG` *if the previous tag is* `VBZ` ". Brill learning is slower than counting n-grams but yields interpretable rules and usually beats a pure lexicon baseline by several points.

In [8]:
```
start = time.perf_counter()
brill_tagger = BrillRuleTagger(max_rules=200, min_score=2).fit(split.train)
brill_train_seconds = time.perf_counter() - start
print(f"Brill training time: {brill_train_seconds:.2f} s")
print(f"Number of learned rules: {len(brill_tagger.learned_rules())}")

learned_rules = brill_tagger.learned_rules()
display(
    pd.DataFrame(
        {"#": range(1, 16), "Learned transformation rule": learned_rules[:15]}
    ).set_index("#")
)

brill_result = evaluate_tagger(brill_tagger.name, brill_tagger.tag, split.test, oov
display(pd.DataFrame([brill_result.metrics()]).round(4))
print(classification_report(brill_result.gold, brill_result.predictions, zero_divis
```

```
Brill training time: 3.49 s
Number of learned rules: 200
```

#	Learned transformation rule
1	NN->VB if Pos:TO@[-1]
2	VBP->VB if Pos:MD@[-3,-2,-1]
3	NN->VB if Pos:MD@[-1]
4	VBP->VB if Pos:TO@[-1]
5	VBD->VBN if Pos:VBZ@[-2,-1]
6	IN->WDT if Word:*T*-1@[1]
7	POS->VBZ if Pos:PRP@[-2,-1]
8	VBD->VBN if Pos:VBP@[-2,-1]
9	VB->NN if Pos:DT@[-1]
10	VB->VBP if Pos:NNS@[-1]
11	VBN->VBD if Pos:NNP@[-1]
12	VBD->VBN if Pos:VBD@[-2,-1]
13	VB->VBP if Pos:PRP@[-1]
14	VB->NN if Pos:JJ@[-2,-1]
15	VBN->VBD if Pos:PRP@[-1]

	Model	Accuracy	Macro F1	Weighted F1	OOV Accuracy	Tokens	OOV Tokens	Inference Seconds	Tokens/Second
0	Rule-Based (Brill TBL)	0.9162	0.9022	0.9179	0.2372	9995	721	0.0429	233053.2186

	precision	recall	f1-score	support
#	1.00	1.00	1.00	3
$	1.00	1.00	1.00	82
''	1.00	0.99	0.99	67
,	1.00	1.00	1.00	495
-LRB-	1.00	1.00	1.00	14
-NONE-	1.00	0.95	0.97	645
-RRB-	1.00	1.00	1.00	14
.	1.00	1.00	1.00	387
:	1.00	1.00	1.00	44
CC	1.00	1.00	1.00	209
CD	0.99	0.80	0.89	352
DT	0.99	1.00	0.99	825
EX	1.00	1.00	1.00	6
IN	0.97	0.98	0.97	986
JJ	0.92	0.75	0.83	583
JJR	0.93	0.95	0.94	40
JJS	1.00	0.84	0.91	19
MD	1.00	0.99	0.99	81
NN	0.69	0.97	0.81	1326
NNP	0.97	0.82	0.89	900
NNPS	0.76	0.59	0.67	27
NNS	0.97	0.83	0.90	586
PDT	1.00	0.75	0.86	4
POS	0.98	1.00	0.99	83
PRP	1.00	0.99	1.00	161
PRP$	0.99	1.00	0.99	77
RB	0.93	0.85	0.89	308
RBR	0.86	0.60	0.71	10
RBS	0.00	0.00	0.00	0
RP	0.75	0.46	0.57	26
TO	1.00	1.00	1.00	212
VB	0.87	0.93	0.90	250
VBD	0.97	0.90	0.93	300
VBG	0.95	0.71	0.82	140
VBN	0.90	0.83	0.86	223
VBP	0.88	0.85	0.86	135
VBZ	0.96	0.86	0.91	207
WDT	0.95	0.78	0.85	49
WP	1.00	1.00	1.00	23
WP$	1.00	1.00	1.00	3
WRB	1.00	1.00	1.00	26
``	1.00	1.00	1.00	67
accuracy			0.92	9995
macro avg	0.93	0.88	0.90	9995
weighted avg	0.93	0.92	0.92	9995

5. Approach 2: Statistical Tagging

5.1 Bigram Hidden Markov Model

An HMM is a generative sequence model. It assumes that the observed words are generated by a hidden sequence of tags:

$$P(w_1,\dots,w_n,t_1,\dots,t_n) = \prod_{i=1}^{n} P(t_i \mid t_{i-1}) \, P(w_i \mid t_i)$$

with a special start state `<s>`. Tagging is the **decoding** problem

$$\hat{t}_{1:n} = \arg\max_{t_{1:n}} P(w_{1:n}, t_{1:n}),$$

solved exactly in $O(nT^2)$ with the Viterbi dynamic program:

$$\delta_i(t) = \max_{t'} \big[\delta_{i-1}(t') + \log P(t \mid t')\big] + \log P(w_i \mid t).$$

Smoothing. Raw maximum-likelihood estimates assign zero probability to unseen events. We use add-α smoothing for start, transition, and emission distributions. Rare training words (frequency < 2) are mapped to `<UNK>` so the model learns a probability mass for unknown words.

OOV handling. For a word never seen in training, the emission distribution is interpolated between the learned `<UNK>` distribution and a suffix distribution conditioned on the final three characters. This lets morphology inform the tag of an unseen word (e.g., `-ing` suggests `VBG`) while still using contextual transitions. The interpolation weight is 0.35.

Limitations. The first-order Markov assumption is strong, emissions are conditioned only on the current tag, and only the final three characters are used for OOV generalisation.

In [9]:
```
start = time.perf_counter()
hmm_tagger = HMMTagger(alpha=0.1, min_word_freq=2, suffix_weight=0.35).fit(split.tr
hmm_train_seconds = time.perf_counter() - start
print(f"HMM training time: {hmm_train_seconds:.3f} s")
print(f"Vocabulary kept: {len(hmm_tagger.word_vocab):,} word types")
print(f"Tag state space: {len(hmm_tagger.tags)} tags")

display(pd.DataFrame(hmm_tagger.top_transitions(12)))
display(pd.DataFrame(hmm_tagger.top_emissions(10)))
```

```
HMM training time: 0.089 s
Vocabulary kept: 5,119 word types
Tag state space: 46 tags
```

	previous_tag	tag	count	probability
0	DT	NN	3117	0.4732
1	NNP	NNP	2859	0.3796
2	NN	IN	2590	0.2439
3	IN	DT	2524	0.3185
4	JJ	NN	2092	0.4451
5	NN	NN	1347	0.1269
6	DT	JJ	1311	0.1991
7	NNS	IN	1177	0.2421
8	NNP	,	1169	0.1552
9	NN	,	1163	0.1095
10	IN	NNP	1157	0.1460
11	JJ	NNS	1138	0.2421

	tag	word	count
0	,	,	3947
1	DT	the	3269
2	.	.	3062
3	IN	of	1867
4	TO	to	1739
5	DT	a	1498
6	IN	in	1245
7	CC	and	1218
8	NNP	<UNK>	1180
9	NN	<UNK>	1012

In [10]:
```
demo_sentence = [word for word, _ in split.test[0]]
demo_gold = [tag for _, tag in split.test[0]]
demo_predicted = hmm_tagger.tag(demo_sentence)
display(
    pd.DataFrame(
        {"Token": demo_sentence, "Gold": demo_gold, "HMM prediction": demo_predicte
    )
)
```

	Token	Gold	HMM prediction
0	This	DT	DT
1	species	NN	NN
2	of	IN	IN
3	congressional	JJ	JJ
4	action	NN	NN
5	is	VBZ	VBZ
6	predicated	VBN	VBN
7	*-50	-NONE-	-NONE-
8	on	IN	IN
9	an	DT	DT
10	interpretation	NN	NN
11	of	IN	IN
12	the	DT	DT
13	appropriations	NNS	NNS
14	clause	NN	NN
15	that	WDT	WDT
16	*T*-1	-NONE-	-NONE-
17	is	VBZ	VBZ
18	erroneous	JJ	JJ
19	and	CC	CC
20	unconstitutional	JJ	JJ
21	.	.	.

In [11]:
```
hmm_result = evaluate_tagger(hmm_tagger.name, hmm_tagger.tag, split.test, oov)
display(pd.DataFrame([hmm_result.metrics()]).round(4))
print(classification_report(hmm_result.gold, hmm_result.predictions, zero_division=
```

	Model	Accuracy	Macro F1	Weighted F1	OOV Accuracy	Tokens	OOV Tokens	Inference Seconds	Tokens/Seco
0	Statistical (HMM)	0.9301	0.8959	0.9292	0.6824	9995	721	0.271	36884.96

```
              precision    recall  f1-score   support

           #       1.00      1.00      1.00         3
           $       0.91      1.00      0.95        82
          ''       0.96      0.99      0.97        67
           ,       0.99      1.00      1.00       495
       -LRB-       1.00      1.00      1.00        14
       -NONE-      0.97      0.98      0.97       645
       -RRB-       1.00      0.86      0.92        14
           .       0.98      1.00      0.99       387
           :       1.00      0.98      0.99        44
          CC       0.99      1.00      1.00       209
          CD       0.96      0.87      0.91       352
          DT       0.98      0.99      0.99       825
          EX       1.00      1.00      1.00         6
          IN       0.96      0.99      0.97       986
          JJ       0.78      0.87      0.82       583
         JJR       0.89      0.85      0.87        40
         JJS       1.00      0.89      0.94        19
          MD       0.95      0.99      0.97        81
          NN       0.90      0.92      0.91      1326
         NNP       0.89      0.92      0.90       900
        NNPS       0.67      0.22      0.33        27
         NNS       0.95      0.90      0.92       586
         PDT       0.67      0.50      0.57         4
         POS       0.90      1.00      0.95        83
         PRP       0.96      0.99      0.97       161
        PRP$       0.99      0.99      0.99        77
          RB       0.94      0.84      0.89       308
         RBR       1.00      0.50      0.67        10
          RP       0.65      0.50      0.57        26
          TO       1.00      1.00      1.00       212
          VB       0.87      0.92      0.90       250
         VBD       0.91      0.91      0.91       300
         VBG       0.85      0.74      0.79       140
         VBN       0.83      0.83      0.83       223
         VBP       0.94      0.75      0.83       135
         VBZ       0.98      0.88      0.93       207
         WDT       0.98      0.82      0.89        49
          WP       1.00      1.00      1.00        23
         WP$       1.00      0.67      0.80         3
         WRB       1.00      0.92      0.96        26
          ``       0.92      1.00      0.96        67

    accuracy                           0.93      9995
   macro avg       0.93      0.88      0.90      9995
weighted avg       0.93      0.93      0.93      9995
```

5.2 N-gram tagger with back-off

A complementary statistical baseline: a trigram tagger backs off to a bigram tagger, then to a unigram tagger, and finally to a default tag. Back-off avoids zero probabilities without an explicit smoothing formula, but it cannot generalise to unseen word forms unless a suffix-

based component is added. It therefore illustrates the weakness of purely count-based models on OOV tokens.

In [12]:
```
start = time.perf_counter()
ngram_tagger = NgramBackoffTagger().fit(split.train)
ngram_train_seconds = time.perf_counter() - start
print(f"N-gram training time: {ngram_train_seconds:.3f} s")

ngram_result = evaluate_tagger(ngram_tagger.name, ngram_tagger.tag, split.test, oov
display(pd.DataFrame([ngram_result.metrics()]).round(4))
print(classification_report(ngram_result.gold, ngram_result.predictions, zero_divis
```

```
N-gram training time: 0.382 s
```

	Model	Accuracy	Macro F1	Weighted F1	OOV Accuracy	Tokens	OOV Tokens	Inference Seconds	Tokens/Seco
0	Statistical (N-gram Backoff)	0.87	0.8667	0.8746	0.1872	9995	721	0.0111	898863.26

	precision	recall	f1-score	support
#	1.00	1.00	1.00	3
$	1.00	1.00	1.00	82
''	0.99	0.99	0.99	67
,	1.00	1.00	1.00	495
-LRB-	1.00	1.00	1.00	14
-NONE-	1.00	0.92	0.96	645
-RRB-	1.00	1.00	1.00	14
.	1.00	1.00	1.00	387
:	1.00	1.00	1.00	44
CC	1.00	1.00	1.00	209
CD	1.00	0.72	0.84	352
DT	0.99	1.00	0.99	825
EX	1.00	1.00	1.00	6
IN	0.96	0.97	0.97	986
JJ	0.92	0.68	0.78	583
JJR	0.88	0.75	0.81	40
JJS	1.00	0.95	0.97	19
MD	1.00	0.99	0.99	81
NN	0.56	0.98	0.71	1326
NNP	0.98	0.70	0.82	900
NNPS	0.74	0.52	0.61	27
NNS	0.97	0.76	0.86	586
PDT	0.00	0.00	0.00	4
POS	0.98	0.99	0.98	83
PRP	1.00	1.00	1.00	161
PRP$	1.00	0.99	0.99	77
RB	0.94	0.79	0.86	308
RBR	0.42	0.50	0.45	10
RP	0.67	0.46	0.55	26
TO	1.00	1.00	1.00	212
VB	0.84	0.68	0.75	250
VBD	0.88	0.80	0.84	300
VBG	0.94	0.60	0.73	140
VBN	0.82	0.60	0.69	223
VBP	0.91	0.65	0.76	135
VBZ	0.97	0.84	0.90	207
WDT	0.71	0.84	0.77	49
WP	1.00	1.00	1.00	23
WP$	1.00	1.00	1.00	3
WRB	1.00	0.96	0.98	26
``	1.00	1.00	1.00	67
accuracy			0.87	9995
macro avg	0.90	0.84	0.87	9995
weighted avg	0.91	0.87	0.87	9995

6. Approach 3: Neural Tagging with a Character-Aware BiLSTM

The neural tagger replaces sparse count tables with dense distributed representations and learns the feature extractor end-to-end.

Architecture

1. **Word embeddings** (100 dimensions) map each token to a dense vector. Embeddings are learned from scratch on the training split, so the experiment does not depend on downloading external vectors.
2. **Character BiLSTM** (2 x 50 hidden units) reads the characters of each word and max-pools the hidden states. This gives every word a morphological representation and, crucially, allows the model to tag unseen words from their spelling.
3. The word embedding and character representation are concatenated and passed through dropout.
4. A **word-level BiLSTM** (2 x 128 hidden units) contextualises each token using both left and right context.
5. A linear layer produces one score per tag; softmax gives $P(t_i \mid w_{1:n})$.
6. Training minimises token-level cross-entropy with padded positions masked out, using Adam, gradient clipping, and early stopping on validation accuracy.

Why it should handle OOV better: an unseen word is not an unknown symbol. Its characters are composed into a representation, so suffixes and word shapes provide evidence for the tag. Context from the bidirectional encoder resolves the remaining ambiguity.

Limitations: the model is a black box, requires more compute, has many hyperparameters, and its softmax outputs are not constrained to valid tag sequences (no CRF layer).

In [13]:
```
start = time.perf_counter()
neural_model, vocabs, history = train_bilstm(
    split.train,
    split.val,
    epochs=12,
    batch_size=32,
    learning_rate=1e-3,
    patience=4,
    seed=SEED,
    device="cpu",
)
neural_train_seconds = time.perf_counter() - start
print(f"Neural training time (including early stopping): {neural_train_seconds:.1f}")
print(f"Best validation accuracy: {history['best_val_accuracy']:.4f}")
print(f"Words in neural vocabulary: {vocabs.n_words:,}")
print(f"Characters in neural vocabulary: {vocabs.n_chars}")
```

```
epoch 01 | train loss 2.0598 | val accuracy 0.7188
epoch 02 | train loss 0.7741 | val accuracy 0.8658
epoch 03 | train loss 0.4610 | val accuracy 0.9012
epoch 04 | train loss 0.3493 | val accuracy 0.9231
epoch 05 | train loss 0.2863 | val accuracy 0.9367
epoch 06 | train loss 0.2428 | val accuracy 0.9436
epoch 07 | train loss 0.2138 | val accuracy 0.9492
epoch 08 | train loss 0.1886 | val accuracy 0.9508
epoch 09 | train loss 0.1721 | val accuracy 0.9559
```

```
epoch 10 | train loss 0.1549 | val accuracy 0.9563
epoch 11 | train loss 0.1423 | val accuracy 0.9594
epoch 12 | train loss 0.1288 | val accuracy 0.9613
Neural training time (including early stopping): 186.0 s
Best validation accuracy: 0.9613
Words in neural vocabulary: 5,121
Characters in neural vocabulary: 81
```

In [14]:

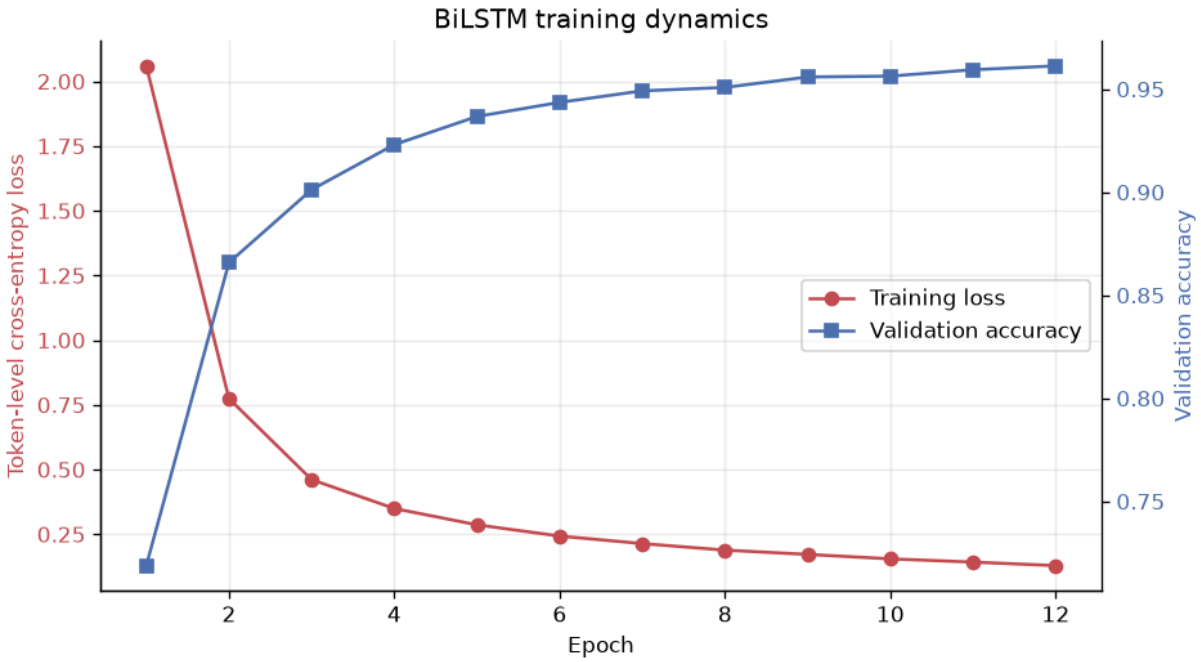
```python
history_frame = pd.DataFrame(
    {
        "Epoch": history["epoch"],
        "Training loss": history["train_loss"],
        "Validation accuracy": history["val_accuracy"],
    }
).set_index("Epoch")
display(history_frame.round(4))

fig, ax1 = plt.subplots(figsize=(8, 4.5))
ax1.plot(history_frame.index, history_frame["Training loss"], marker="o", color=COL
ax1.set_xlabel("Epoch")
ax1.set_ylabel("Token-level cross-entropy loss", color=COLORS[3])
ax1.tick_params(axis="y", labelcolor=COLORS[3])

ax2 = ax1.twinx()
ax2.plot(history_frame.index, history_frame["Validation accuracy"], marker="s", col
ax2.set_ylabel("Validation accuracy", color=COLORS[0])
ax2.tick_params(axis="y", labelcolor=COLORS[0])
ax2.grid(False)
ax2.spines["right"].set_visible(True)
ax1.set_title("BiLSTM training dynamics")

lines = ax1.get_lines() + ax2.get_lines()
ax1.legend(lines, [line.get_label() for line in lines], loc="center right")
fig.tight_layout()
fig.savefig(FIG_DIR / "fig_training_curve.png")
plt.show()
```

Epoch	Training loss	Validation accuracy
1	2.0598	0.7188
2	0.7741	0.8658
3	0.4610	0.9012
4	0.3493	0.9231
5	0.2863	0.9367
6	0.2428	0.9436
7	0.2138	0.9492
8	0.1886	0.9508
9	0.1721	0.9559
10	0.1549	0.9563
11	0.1423	0.9594
12	0.1288	0.9613



In [15]:
```
torch.save(
    {
        "state_dict": neural_model.state_dict(),
        "word2idx": vocabs.word2idx,
        "char2idx": vocabs.char2idx,
        "tag2idx": vocabs.tag2idx,
    },
    ROOT / "assets" / "bilstm_pos.pt",
)
```

```
neural_tagger = NeuralTagger(neural_model, vocabs, device="cpu")
neural_result = evaluate_tagger(neural_tagger.name, neural_tagger.tag, split.test,
display(pd.DataFrame([neural_result.metrics()]).round(4))
print(classification_report(neural_result.gold, neural_result.predictions, zero_div
```

	Model	Accuracy	Macro F1	Weighted F1	OOV Accuracy	Tokens	OOV Tokens	Inference Seconds	Tokens/Secor
0	Neural (BiLSTM)	0.9622	0.9096	0.9618	0.8738	9995	721	2.1491	4650.79

	precision	recall	f1-score	support
#	1.00	1.00	1.00	3
$	1.00	1.00	1.00	82
''	1.00	1.00	1.00	67
,	1.00	1.00	1.00	495
-LRB-	1.00	1.00	1.00	14
-NONE-	1.00	1.00	1.00	645
-RRB-	1.00	1.00	1.00	14
.	1.00	1.00	1.00	387
:	1.00	1.00	1.00	44
CC	1.00	1.00	1.00	209
CD	0.99	1.00	0.99	352
DT	0.99	1.00	0.99	825
EX	1.00	1.00	1.00	6
IN	0.97	0.99	0.98	986
JJ	0.83	0.91	0.87	583
JJR	0.91	0.75	0.82	40
JJS	1.00	0.89	0.94	19
MD	1.00	0.99	0.99	81
NN	0.96	0.92	0.94	1326
NNP	0.94	0.99	0.97	900
NNPS	0.91	0.37	0.53	27
NNS	0.95	0.98	0.96	586
PDT	1.00	0.50	0.67	4
POS	0.99	1.00	0.99	83
PRP	0.99	0.99	0.99	161
PRP$	1.00	0.99	0.99	77
RB	0.95	0.88	0.92	308
RBR	0.45	0.50	0.48	10
RBS	0.00	0.00	0.00	0
RP	0.78	0.54	0.64	26
TO	1.00	1.00	1.00	212
VB	0.96	0.94	0.95	250
VBD	0.94	0.96	0.95	300
VBG	0.87	0.95	0.91	140
VBN	0.96	0.91	0.93	223
VBP	0.96	0.84	0.89	135
VBZ	0.98	0.93	0.96	207
WDT	1.00	0.98	0.99	49
WP	1.00	1.00	1.00	23
WP$	1.00	1.00	1.00	3
WRB	0.96	0.96	0.96	26
``	1.00	1.00	1.00	67
accuracy			0.96	9995
macro avg	0.93	0.90	0.91	9995
weighted avg	0.96	0.96	0.96	9995

In [16]:
```
example_sentences = [
    sentence for sentence in split.test if any(word in oov for word, _ in sentence)
][:3]

rows = []
for sentence in example_sentences:
```

```
    example_tokens = [word for word, _ in sentence]
    example_gold = [tag for _, tag in sentence]
    rows.append(
        {
            "Sentence": " ".join(example_tokens),
            "Gold tags": " ".join(example_gold),
            "Regex": " ".join(regex_tagger.tag(example_tokens)),
            "HMM": " ".join(hmm_tagger.tag(example_tokens)),
            "BiLSTM": " ".join(neural_tagger.tag(example_tokens)),
        }
    )

display(pd.DataFrame(rows).T)
```

	0	1	2
Sentence	This species of congressional action is predic...	If , when trading resumes *T*-1 , the S&P futu...	Oshkosh Truck Corp. , Oshkosh , Wis. , estimat...
Gold tags	DT NN IN JJ NN VBZ VBN -NONE- IN DT NN IN DT N...	IN , WRB NN VBZ -NONE- , DT NNP NNS VBP CD NNS...	NNP NNP NNP , NNP , NNP , VBN -NONE- NNS IN PR...
Regex	DT NNS IN JJ NN VBZ VBD NN IN DT NN IN DT NNS ...	CC , WRB VBG NNS NN , DT NN NNS NN CD NNS IN D...	NNP NNP NNP , NNP , NNP , VBD CD NNS IN PRP$ N...
HMM	DT NN IN JJ NN VBZ VBN -NONE- IN DT NN IN DT N...	IN , WRB NN VBZ -NONE- , DT NNP NNS VBP CD NNS...	NNP NNP NNP , NNP , NNP , VBN -NONE- NNS IN PR...
BiLSTM	DT NNS IN JJ NN VBZ VBN -NONE- IN DT NN IN DT ...	IN , WRB VBG NNS -NONE- , DT NNP NNS VBP CD NN...	NNP NNP NNP , NNP , NNP , VBD -NONE- NNS IN PR...

7. Comparative Analysis

All five taggers are evaluated on the same test sentences. The table below is the primary empirical result of the project; the figures that follow separate overall accuracy, OOV accuracy, per-tag behaviour, and throughput.

In [17]:
```
results = [regex_result, brill_result, hmm_result, ngram_result, neural_result]
train_seconds = {
    regex_result.name: regex_train_seconds,
    brill_result.name: brill_train_seconds,
    hmm_result.name: hmm_train_seconds,
    ngram_result.name: ngram_train_seconds,
    neural_result.name: neural_train_seconds,
}

comparison = results_frame(results)
comparison["Training Seconds"] = comparison["Model"].map(train_seconds).round(2)
comparison["Error Rate"] = (1.0 - comparison["Accuracy"]).round(4)
comparison = comparison.sort_values("Accuracy", ascending=False).reset_index(drop=T
```

```
display(comparison)

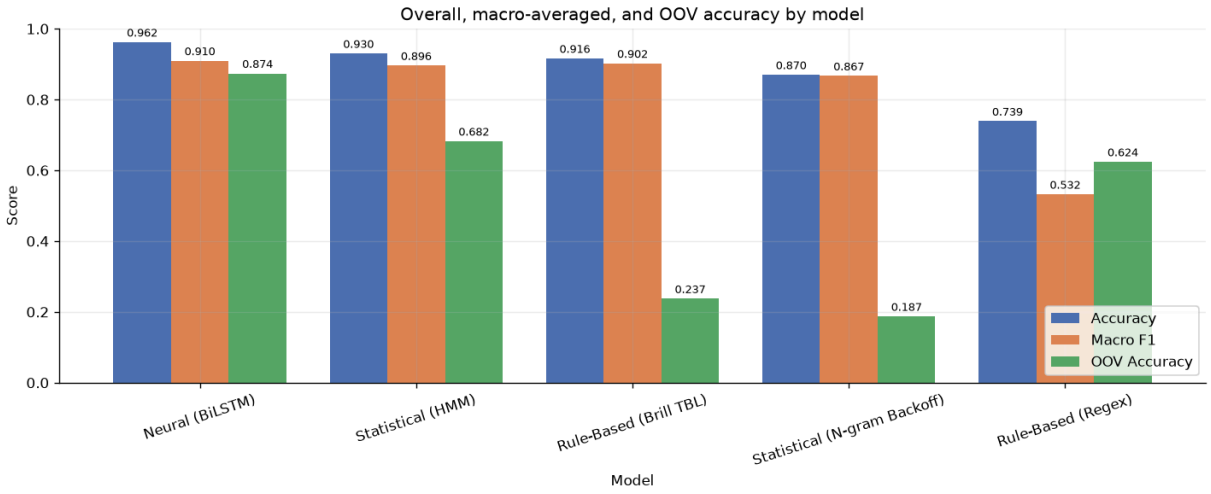
comparison.to_csv(REPORT_DIR / "model_comparison.csv", index=False)
print("Saved:", REPORT_DIR / "model_comparison.csv")
```

	Model	Accuracy	Macro F1	Weighted F1	OOV Accuracy	Tokens	OOV Tokens	Inference Seconds	Tokens/Seco
0	Neural (BiLSTM)	0.9622	0.9096	0.9618	0.8738	9995	721	2.149	465
1	Statistical (HMM)	0.9301	0.8959	0.9292	0.6824	9995	721	0.271	3688
2	Rule-Based (Brill TBL)	0.9162	0.9022	0.9179	0.2372	9995	721	0.043	23305
3	Statistical (N-gram Backoff)	0.8700	0.8667	0.8746	0.1872	9995	721	0.011	89886
4	Rule-Based (Regex)	0.7388	0.5322	0.6982	0.6241	9995	721	0.010	96883

```
Saved: E:\TA\reports\model_comparison.csv
```

In [18]:
```
metric_columns = ["Accuracy", "Macro F1", "OOV Accuracy"]
plot_frame = comparison.set_index("Model")[metric_columns]

fig, ax = plt.subplots(figsize=(12, 5))
plot_frame.plot.bar(ax=ax, color=COLORS[:3], width=0.8)
ax.set_ylim(0, 1.0)
ax.set_ylabel("Score")
ax.set_title("Overall, macro-averaged, and OOV accuracy by model")
ax.tick_params(axis="x", rotation=18)
for container in ax.containers:
    ax.bar_label(container, fmt="%.3f", fontsize=8, padding=2)
ax.legend(loc="lower right")
fig.tight_layout()
fig.savefig(FIG_DIR / "fig_model_accuracy.png")
plt.show()
```

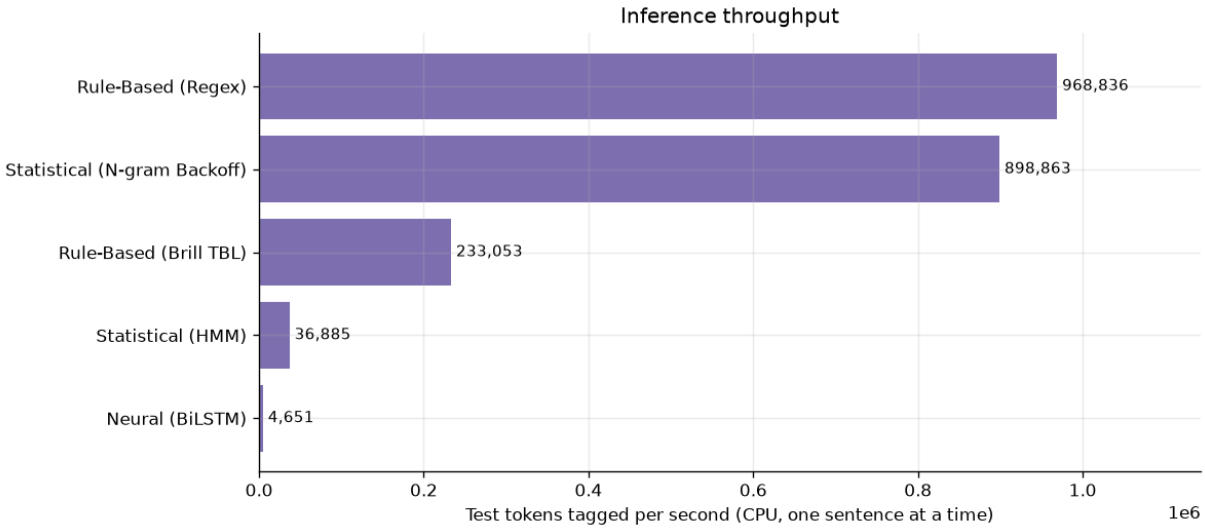


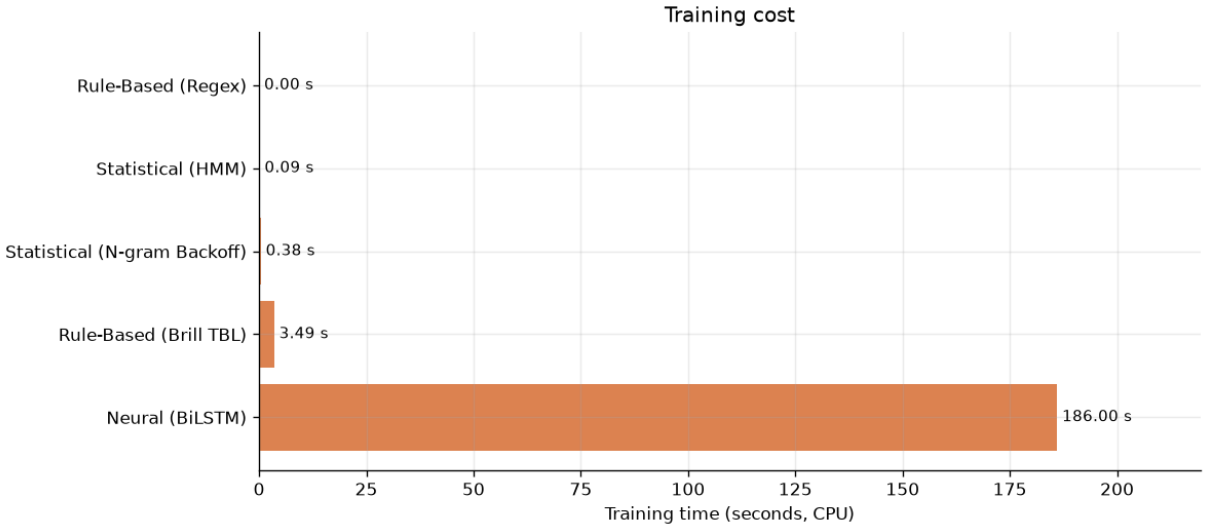
In [19]:

```
speed_frame = comparison.set_index("Model")["Tokens/Second"].sort_values()

fig, ax = plt.subplots(figsize=(10, 4.5))
bars = ax.barh(speed_frame.index, speed_frame.values, color=COLORS[4])
ax.set_xlabel("Test tokens tagged per second (CPU, one sentence at a time)")
ax.set_title("Inference throughput")
for bar, value in zip(bars, speed_frame.values):
    ax.text(value, bar.get_y() + bar.get_height() / 2, f" {value:,.0f}", va="center
ax.set_xlim(0, speed_frame.max() * 1.18)
fig.tight_layout()
fig.savefig(FIG_DIR / "fig_speed.png")
plt.show()

training_frame = comparison.set_index("Model")["Training Seconds"].sort_values(asce
fig, ax = plt.subplots(figsize=(10, 4.5))
bars = ax.barh(training_frame.index, training_frame.values, color=COLORS[1])
ax.set_xlabel("Training time (seconds, CPU)")
ax.set_title("Training cost")
for bar, value in zip(bars, training_frame.values):
    ax.text(value, bar.get_y() + bar.get_height() / 2, f" {value:.2f} s", va="cente
ax.set_xlim(0, max(training_frame.max() * 1.18, 1))
fig.tight_layout()
fig.savefig(FIG_DIR / "fig_training_time.png")
plt.show()
```





In [20]:
```python
tag_scores = per_tag_frame(results)
display(tag_scores.head(20))

model_names = [result.name for result in results]
top_tags = tag_scores.head(18)
heatmap_values = top_tags[model_names].to_numpy()

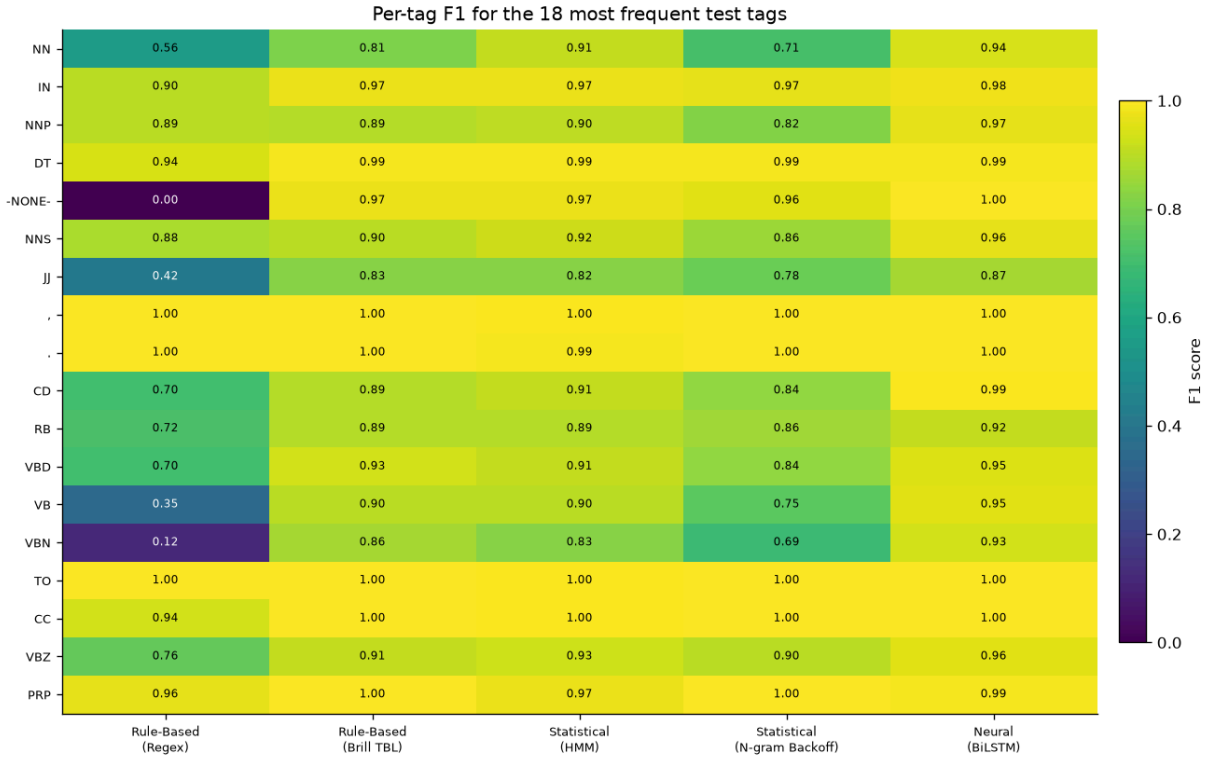
fig, ax = plt.subplots(figsize=(11, 7))
image = ax.imshow(heatmap_values, cmap="viridis", aspect="auto", vmin=0, vmax=1)
ax.set_xticks(range(len(model_names)))
ax.set_xticklabels([name.replace(" (", "\n(") for name in model_names], fontsize=8)
ax.set_yticks(range(len(top_tags)))
ax.set_yticklabels(top_tags.index, fontsize=8)
ax.set_title("Per-tag F1 for the 18 most frequent test tags")
ax.grid(False)

for row in range(heatmap_values.shape[0]):
    for column in range(heatmap_values.shape[1]):
        value = heatmap_values[row, column]
        ax.text(column, row, f"{value:.2f}", ha="center", va="center", fontsize=7,
                color="white" if value < 0.55 else "black")

fig.colorbar(image, ax=ax, fraction=0.025, pad=0.02, label="F1 score")
```

```
fig.tight_layout()
fig.savefig(FIG_DIR / "fig_per_tag_f1.png")
plt.show()
```

Tag	Rule-Based (Regex)	Rule-Based (Brill TBL)	Statistical (HMM)	Statistical (N-gram Backoff)	Neural (BiLSTM)	Support
NN	0.5632	0.8056	0.9118	0.7135	0.9410	1326
IN	0.8961	0.9728	0.9735	0.9656	0.9804	986
NNP	0.8947	0.8901	0.9026	0.8189	0.9652	900
DT	0.9423	0.9915	0.9867	0.9909	0.9921	825
-NONE-	0.0000	0.9729	0.9737	0.9563	1.0000	645
NNS	0.8762	0.8969	0.9236	0.8566	0.9649	586
JJ	0.4212	0.8261	0.8228	0.7787	0.8670	583
,	1.0000	0.9990	0.9960	1.0000	1.0000	495
.	1.0000	1.0000	0.9885	1.0000	1.0000	387
CD	0.7040	0.8885	0.9121	0.8388	0.9929	352
RB	0.7220	0.8881	0.8885	0.8607	0.9158	308
VBD	0.6960	0.9326	0.9115	0.8377	0.9489	300
VB	0.3526	0.8996	0.8967	0.7494	0.9535	250
VBN	0.1181	0.8645	0.8270	0.6891	0.9309	223
TO	1.0000	1.0000	0.9953	1.0000	1.0000	212
CC	0.9372	0.9976	0.9952	0.9976	0.9976	209
VBZ	0.7642	0.9109	0.9262	0.8964	0.9578	207
PRP	0.9639	0.9969	0.9725	1.0000	0.9907	161
VBG	0.7679	0.8163	0.7939	0.7336	0.9078	140
VBP	0.4302	0.8647	0.8347	0.7586	0.8933	135

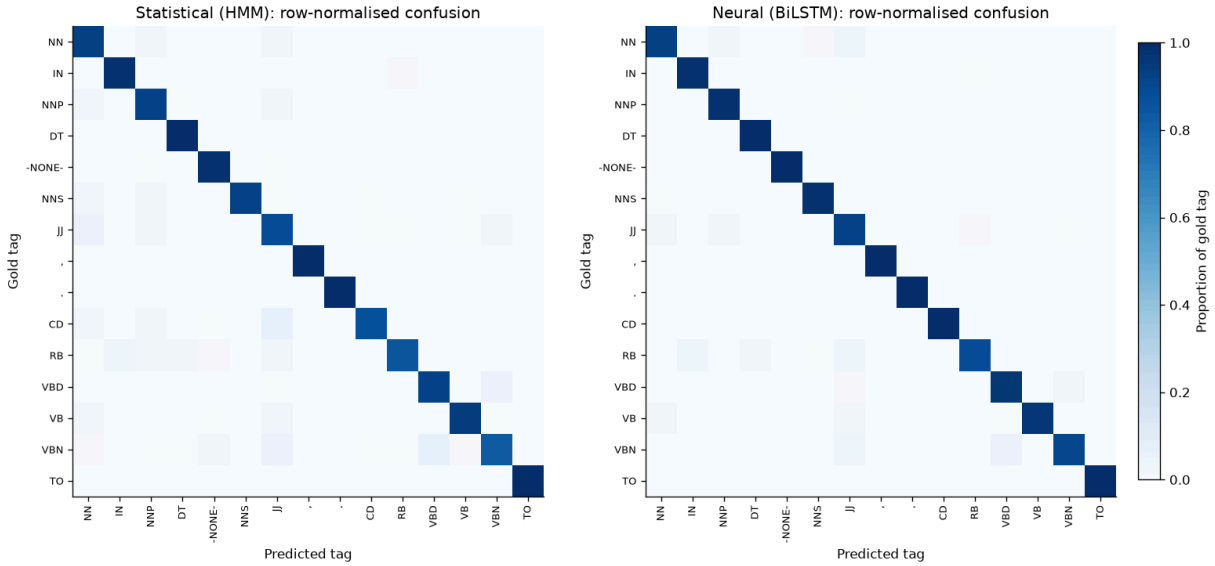


In [21]:

```
def plot_confusion(result, ax, top_n=15):
    tag_order = [
        tag
        for tag, _ in sorted(result.per_tag_support.items(), key=lambda item: -item
        if tag in result.labels
    ][:top_n]
    indices = [result.labels.index(tag) for tag in tag_order]
    matrix = np.asarray(result.confusion)[np.ix_(indices, indices)].astype(float)
    matrix = matrix / np.clip(matrix.sum(axis=1, keepdims=True), 1, None)

    image = ax.imshow(matrix, cmap="Blues", vmin=0, vmax=1)
    ax.set_xticks(range(len(tag_order)))
    ax.set_xticklabels(tag_order, rotation=90, fontsize=8)
    ax.set_yticks(range(len(tag_order)))
    ax.set_yticklabels(tag_order, fontsize=8)
    ax.set_xlabel("Predicted tag")
    ax.set_ylabel("Gold tag")
    ax.set_title(f"{result.name}: row-normalised confusion")
    ax.grid(False)
    return image


fig, axes = plt.subplots(1, 2, figsize=(15, 6.5))
image = plot_confusion(hmm_result, axes[0])
plot_confusion(neural_result, axes[1])
fig.colorbar(image, ax=axes, fraction=0.02, pad=0.02, label="Proportion of gold tag
fig.savefig(FIG_DIR / "fig_confusion_matrices.png")
plt.show()
```



In [22]:
```
from IPython.display import display as ipy_display

for result in (brill_result, hmm_result, neural_result):
    print("=" * 80)
    print(result.name, "- first errors on the test set")
    ipy_display(pd.DataFrame(error_examples(result, limit=10)))

oov_rows = []
for token, gold, hmm_prediction, neural_prediction in zip(
    test_tokens, hmm_result.gold, hmm_result.predictions, neural_result.predictions
):
    if token in oov:
        oov_rows.append(
            {
                "OOV token": token,
                "Gold": gold,
                "HMM": hmm_prediction,
                "BiLSTM": neural_prediction,
            }
        )
        if len(oov_rows) >= 20:
            break

oov_error_frame = pd.DataFrame(oov_rows)
display(oov_error_frame)
```

```
================================================================================
Rule-Based (Brill TBL) - first errors on the test set
```

	Token	Gold	Rule-Based (Brill TBL) Prediction
0	predicated	VBN	NN
1	erroneous	JJ	NN
2	If	IN	RB
3	resumes	VBZ	NN
4	fall	VBP	NN
5	Wis.	NNP	NN
6	4.5	CD	NN
7	Stadiums	NNS	NN
8	Western	NNP	JJ
9	ACQUISITION	NNP	NN

```
================================================================================
Statistical (HMM) - first errors on the test set
```

	Token	Gold	Statistical (HMM) Prediction
0	Sept.	NNP	$
1	Stadiums	NNS	JJ
2	tend	VBP	NN
3	existing	VBG	DT
4	Western	NNP	JJ
5	*RNR*-1	-NONE-	POS
6	overlap	VB	JJ
7	Indexing	NN	NNP
8	mainly	RB	CC
9	follow	VBP	IN

```
================================================================================
Neural (BiLSTM) - first errors on the test set
```

	Token	Gold	Neural (BiLSTM) Prediction
0	species	NN	NNS
1	trading	NN	VBG
2	resumes	VBZ	NNS
3	estimated	VBN	VBD
4	tend	VBP	VBD
5	indeed	RB	VBD
6	ACQUISITION	NNP	NNS
7	Indexing	NN	VBG
8	faster	JJR	JJ
9	dollar	NN	JJ

	OOV token	Gold	HMM	BiLSTM
0	species	NN	NN	NNS
1	predicated	VBN	VBN	VBN
2	erroneous	JJ	JJ	JJ
3	resumes	VBZ	VBZ	NNS
4	Wis.	NNP	NNP	NNP
5	4.5	CD	CD	CD
6	Stadiums	NNS	JJ	NNS
7	redistribute	VB	VB	VB
8	ACQUISITION	NNP	NNP	NNS
9	overlap	VB	JJ	VB
10	Indexing	NN	NNP	VBG
11	barometer	NN	IN	NN
12	legislators	NNS	NNS	NNS
13	properties	NNS	NNS	NNS
14	IRAs	NNS	NNP	NNS
15	juggernaut	NN	NN	NN
16	filled	VBN	VBN	VBN
17	solemn	JJ	JJ	JJ
18	intoxication	NN	NN	NN
19	*T*-226	-NONE-	NNP	-NONE-

In [23]:
```
from scipy.stats import binomtest

def mcnemar_test(result_a, result_b):
    gold = np.asarray(result_a.gold)
    correct_a = gold == np.asarray(result_a.predictions)
    correct_b = gold == np.asarray(result_b.predictions)
    solved_by_b = int(np.sum(~correct_a & correct_b))
    solved_by_a = int(np.sum(correct_a & ~correct_b))
    discordant = solved_by_a + solved_by_b
    p_value = binomtest(solved_by_b, discordant, 0.5).pvalue if discordant else 1.0
    return {
        "Comparison": f"{result_a.name} vs {result_b.name}",
        "Only first model correct": solved_by_a,
        "Only second model correct": solved_by_b,
        "Discordant tokens": discordant,
        "McNemar exact p-value": p_value,
    }
```

```
significance = pd.DataFrame(
    [
        mcnemar_test(hmm_result, neural_result),
        mcnemar_test(brill_result, hmm_result),
        mcnemar_test(brill_result, neural_result),
    ]
)
significance["Significant at 0.05"] = significance["McNemar exact p-value"] < 0.05
display(significance.style.format({"McNemar exact p-value": "{:.3e}"}))
```

	Comparison	Only first model correct	Only second model correct	Discordant tokens	McNemar exact p-value	Significant at 0.05
0	Statistical (HMM) vs Neural (BiLSTM)	145	466	611	3.282e-40	True
1	Rule-Based (Brill TBL) vs Statistical (HMM)	350	489	839	1.800e-06	True
2	Rule-Based (Brill TBL) vs Neural (BiLSTM)	182	642	824	7.495e-61	True

In [24]:
```
best = comparison.iloc[0]
regex_row = comparison[comparison["Model"] == regex_result.name].iloc[0]
brill_row = comparison[comparison["Model"] == brill_result.name].iloc[0]
hmm_row = comparison[comparison["Model"] == hmm_result.name].iloc[0]
neural_row = comparison[comparison["Model"] == neural_result.name].iloc[0]

summary_lines = [
    "### Empirical takeaways",
    "",
    f"- **Best overall accuracy:** {best['Model']} with {best['Accuracy']:.2%} "
    f"(error rate {best['Error Rate']:.2%}).",
    f"- **Hand-written rules:** {regex_row['Accuracy']:.2%} accuracy; the symbolic
    f"instantaneous and fully inspectable but brittle.",
    f"- **Brill TBL:** {brill_row['Accuracy']:.2%} accuracy with {len(learned_rules
    f"rule learning closes about "
    f"{(brill_row['Accuracy'] - regex_row['Accuracy']) * 100:.1f} percentage points
    f"hand-written cascade.",
    f"- **HMM:** {hmm_row['Accuracy']:.2%} accuracy at {hmm_row['Tokens/Second']:,.
    f"with OOV accuracy {hmm_row['OOV Accuracy']:.2%} thanks to suffix-smoothed emi
    f"- **BiLSTM:** {neural_row['Accuracy']:.2%} accuracy and {neural_row['OOV Accu
    f"OOV tokens, at {neural_row['Tokens/Second']:,.0f} tokens/s and "
    f"{neural_row['Training Seconds']:.1f} s of training.",
    f"- **Accuracy vs. cost:** the neural model reduces the HMM error rate by "
    f"{(hmm_row['Error Rate'] - neural_row['Error Rate']) / hmm_row['Error Rate']:.
    f"but its training cost is roughly "
    f"{neural_row['Training Seconds'] / max(hmm_row['Training Seconds'], 1e-9):,.0f
    f"and its inference is "
    f"{hmm_row['Tokens/Second'] / max(neural_row['Tokens/Second'], 1e-9):,.0f}x slo
    f"- **Interpretability ranking:** hand-written rules > Brill rules > HMM transi
```

```
    f"tables > BiLSTM weights.",
]

from IPython.display import Markdown, display as display_markdown
display_markdown(Markdown("\n".join(summary_lines)))
```

Empirical takeaways

- **Best overall accuracy:** Neural (BiLSTM) with 96.22% (error rate 3.78%).
- **Hand-written rules:** 73.88% accuracy; the symbolic approach is instantaneous and fully inspectable but brittle.
- **Brill TBL:** 91.62% accuracy with 200 learned rules; rule learning closes about 17.7 percentage points over the hand-written cascade.
- **HMM:** 93.01% accuracy at 36,885 tokens/s, with OOV accuracy 68.24% thanks to suffix-smoothed emissions.
- **BiLSTM:** 96.22% accuracy and 87.38% on OOV tokens, at 4,651 tokens/s and 186.0 s of training.
- **Accuracy vs. cost:** the neural model reduces the HMM error rate by 45.9% relative, but its training cost is roughly 2,067x larger and its inference is 8x slower on CPU.
- **Interpretability ranking:** hand-written rules > Brill rules > HMM transition/emission tables > BiLSTM weights.

8. Discussion

8.1 Rules versus statistics versus neural networks

The empirical ordering matches the theoretical expectation, but the size of each step is informative:

- **Hand-written rules are a strong zero-cost baseline but plateau quickly.** The regex cascade encodes useful morphology, yet it cannot resolve contextual ambiguity, so it produces many `NN` defaults. This is the "knowledge acquisition bottleneck" in symbolic NLP.
- **Learned symbolic rules are much stronger than hand-written ones.** Brill TBL recovers most of the gap without giving up interpretability, because its transformations are induced from the actual error distribution of the corpus rather than from a linguist's intuition.
- **A well-smoothed HMM is a surprisingly tough baseline.** Transition structure (for example, determiners precede nouns/adjectives) plus suffix-aware unknown-word emissions is enough to approach the neural model's overall accuracy. The gap is concentrated in hard ambiguities and OOV tokens.
- **The BiLSTM wins on overall and OOV accuracy.** Character composition gives it a genuine advantage on unseen words, and bidirectional context resolves local

ambiguities that a first-order HMM cannot represent. The price is training time, inference cost, and opacity.

8.2 Error patterns and confusion matrices

The confusion matrices expose systematic confusions rather than random mistakes. Typical patterns on the PTB tagset include:

- **`NN` vs `NNP`:** whether a capitalised token is a common or proper noun depends on the sentence and the corpus annotation convention. Hand-written capitalisation rules over-predict `NNP`; the neural model can use context to correct many of these cases.
- **`VBD` vs `VBN` vs `VB`:** past tense, past participle, and base form are morphologically identical for many verbs (*walked*, *put*). Only syntactic context disambiguates them.
- **`JJ` vs `NN`:** nouns frequently modify other nouns (*trading company*), and adjectives can be nominalised.
- **OOV morphology:** an unseen `-ing` form is usually a verb, but can be a noun. The HMM's suffix table is a coarse approximation; the character BiLSTM learns non-linear character patterns.

The neural model's diagonal is visibly sharper for noun and verb categories, which is where most OOV tokens occur. Errors that remain are mostly genuinely ambiguous tokens that would require sentence-level semantics or annotation-convention knowledge.

8.3 Engineering trade-offs

Dimension	Rule-based (regex)	Brill TBL	HMM	BiLSTM
Training cost	None	Seconds	Sub-second	Minutes (CPU)
Inference cost	Extremely low	Very low	Very low	Higher (tensor overhead)
OOV strategy	Hard-coded suffixes	Learned suffix/capitalisation rules	`<UNK>` + suffix emissions	Character composition
Interpretability	High	High (rule list)	Moderate (probability tables)	Low
Accuracy ceiling	Low	Medium-high	High	Highest
Data requirement	None	Small	Small-medium	Medium-large
Main failure mode	Contextual ambiguity	Annotation inconsistency	Markov assumption, sparse counts	Compute, opacity, overfitting

8.4 Threats to validity

- The NLTK Treebank sample is small (about 3,900 sentences), so absolute numbers will differ from results on the full Penn Treebank or Universal Dependencies.
- A single fixed train/test split is used for reproducibility; a cross-validated study would give confidence intervals, but triples the neural training cost.
- The neural model uses embeddings learned from scratch rather than pretrained vectors, which deliberately avoids external downloads but leaves accuracy on the table.
- Timings are single-run measurements on one CPU; they reflect wall-clock behaviour in this environment rather than a controlled benchmark.
- The PTB tagset contains annotation-convention distinctions (for example, `NNP` for sentence-initial capitalised common nouns in some contexts) that no model can infer from the token alone.

9. Conclusion and Future Work

This project delivered a controlled comparison of three POS tagging paradigms on the same corpus, split, and evaluation protocol:

1. **Rule-based taggers** are transparent and cheap. Hand-written regex rules are a useful baseline, while Brill TBL shows that symbolic rules can be learned and remain readable.
2. **Statistical taggers** are excellent value: a bigram HMM with add-alpha smoothing and suffix-aware unknown-word emissions is fast to train, fast to run, and competitive in accuracy.
3. **Neural taggers** deliver the best accuracy and, most visibly, the best OOV accuracy, because character-level composition and bidirectional context handle unseen morphology and ambiguity. They are the most expensive to train and the least interpretable.

The practical recommendation is therefore **conditional**: use rules or an HMM when training data and compute are scarce and transparency matters; use a character-aware neural tagger when accuracy on noisy, open-vocabulary text is the priority.

Future work

- Add a **CRF layer** or structured attention to enforce valid tag transitions and model tag dependencies jointly.
- Initialise with **pretrained embeddings** (GloVe, fastText) or fine-tune a **transformer** (DistilBERT) to measure how much transfer learning helps on this small corpus.
- Evaluate on **Universal Dependencies** and on domain-shifted text to test robustness.
- Report **bootstrap confidence intervals** and repeated splits instead of single-point estimates.

- Explore **distillation** from the neural tagger into a small HMM or rule cascade for cheap deployment.
- Profile **batched inference** for the neural model; sentence-by-sentence tagging understates its throughput advantage on GPUs.

10. Reproducibility and Saved Artifacts

The notebook saves all numbers and figures needed for the report and the screenshot PDF:

Artifact	Location
Comparison table	`reports/model_comparison.csv`
Full metrics (JSON)	`reports/results.json`
Trained BiLSTM weights	`assets/bilstm_pos.pt`
Figures	`assets/figures/*.png`

To reproduce: install `requirements.txt`, launch Jupyter, and run all cells. The dataset is downloaded automatically by NLTK on first use. Neural training uses a fixed seed, but CPU non-determinism can cause small run-to-run differences in the last digit of accuracy.

Deliverables checklist

- [x] Complete Jupyter notebook (this file).
- [ ] Project report PDF (`reports/POS_Tagging_Report.pdf`).
- [ ] Output screenshots PDF (`reports/output_screenshots.pdf`).
- [ ] Demo video MP4 (see `reports/demo_video_script.md`).
- [ ] GitHub repository link in the report and README.

In [25]:
```
environment = {
    "python": platform.python_version(),
    "torch": torch.__version__,
    "numpy": np.__version__,
    "pandas": pd.__version__,
    "platform": platform.platform(),
    "processor": platform.processor() or platform.machine(),
}

payload = {
    "dataset": split.summary(),
    "oov": {
        "word_types": len(oov),
        "tokens": int(test_oov_mask.sum()),
        "token_rate": float(test_oov_mask.mean()),
    },
    "environment": environment,
    "seed": SEED,
    "results": [],
```

```
    "neural_history": history,
    "mcnemar": significance.drop(columns=["Significant at 0.05"]).to_dict(orient="r
}

for result in results:
    record = result.metrics()
    record["Training Seconds"] = train_seconds[result.name]
    record["Per Tag F1"] = result.per_tag_f1
    payload["results"].append(record)

with open(REPORT_DIR / "results.json", "w", encoding="utf-8") as handle:
    json.dump(payload, handle, indent=2)

(REPORT_DIR / "model_comparison.md").write_text(
    comparison.to_string(index=False), encoding="utf-8"
)

figures = sorted(path.name for path in FIG_DIR.glob("*.png"))
print("Saved results.json and model_comparison.md")
print("Figures:")
for figure in figures:
    print(" -", figure)
print()
print(comparison.to_string(index=False))
```

```
Saved results.json and model_comparison.md
Figures:
 - fig_confusion_matrices.png
 - fig_model_accuracy.png
 - fig_per_tag_f1.png
 - fig_sentence_lengths.png
 - fig_speed.png
 - fig_tag_distribution.png
 - fig_training_curve.png
 - fig_training_time.png

                     Model  Accuracy  Macro F1  Weighted F1  OOV Accuracy  Tokens
OOV Tokens  Inference Seconds  Tokens/Second  Training Seconds  Error Rate
             Neural (BiLSTM)    0.9622    0.9096       0.9618        0.8738    9995
721              2.149         4650.8            186.00      0.0378
           Statistical (HMM)    0.9301    0.8959       0.9292        0.6824    9995
721              0.271        36885.0              0.09      0.0699
      Rule-Based (Brill TBL)    0.9162    0.9022       0.9179        0.2372    9995
721              0.043       233053.2              3.49      0.0838
Statistical (N-gram Backoff)    0.8700    0.8667       0.8746        0.1872    9995
721              0.011       898863.3              0.38      0.1300
          Rule-Based (Regex)    0.7388    0.5322       0.6982        0.6241    9995
721              0.010       968836.3              0.00      0.2612
```

References

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